

Power market equilibrium considering an EV storage aggregator exposed to marginal prices - A bilevel optimization approach

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ABSTRACT

This paper presents an economic model that aims to evaluate -from the regulatory perspective- the behavior of an aggregator capable of exploiting business opportunities of electric vehicles (EV) in the context of a wholesale electricity market. As a key contribution, we investigate the market equilibrium point when only one firm acts as an EV aggregator taking advantage of the management of the storage assets. To do so, a Stackelberg game is stated as a bi-level optimization problem where the maximization of the EV aggregator profit is regarded as the leader's optimization problem, and the set of hourly economic dispatches and the corresponding locational marginal prices (LMPs) are determined by a benevolent system operator (the follower's optimization problem). The model was applied in the IEEE RTS test system and the Colombian electricity market. The best solution for the EV aggregator does not require a significant share of the existing storage capacity. However, despite the loss of efficiency observed in the short-term energy market, the oligopolistic solution leads to flattening load curves and, therefore, a better use of the transmission and distribution infrastructure in the long term.

1. Introduction

Electric Vehicles (EVs) technology has reached the maturity needed to ensure a slow but constant growth in the mobility market in the near future [1]. Conservative forecasts showed that by 2035, the size of the world's EV fleet might represent around 6% of the total fleet at a global scale, approximately one hundred million units [2]. The International Energy Agency foresees two hundred millions of EVs by 2030 considering climate change mitigation goals, the increasing interest of the users in emerging mobility technologies, and lowered battery costs [3]. Furthermore, there are also more optimistic views like Bloomberg that forecast EV sales surpassing the ones of Internal Combustion Engine vehicles (ICE) by 2038 [4]. Governments are also promoting new mobility policies and cleaner technologies [5]. The electrification of road transportation will require the integration of EVs into a reliable and affordable as well as easy-of-use infrastructure for the supply of energy [6]. The switch from ICE to EV is recognized as a great opportunity to achieve the ongoing environmental goals [7].

The environmental benefits of EVs integration should be weighed with the economic and technical impact upon the traditional electric distribution infrastructure [8]. Some studies have shown that uncoordinated charge of the EVs can reduce the load factor by enlarging the difference between peak and off-peak demand and increasing the

peak-load duration [9,10]. This result would imply higher losses and congestion with additional reinforcement/expansion costs to be paid by all network users. The drawbacks can be mitigated when the majority of EVs are charged at night and early morning hours outside of regular demand peak hours [11]. As a matter of fact, demand peaks are usually supplied by expensive thermal power stations, so flattening the demand profile has inherent economic and technical benefits. Nevertheless, uncoordinated charging of EVs at peak hours can lead to increased fossil-based energy and emissions [12].

The provision of a coordinated charging scheme for the EV fleet will require economic incentives to aggregators and EV owners. Thus, smart grid applications such as Distribution Automation (DA), Advanced Metering Infrastructure (AMI) and Smart Metering (SM) capabilities must be available [13]. Indeed, when the EV's hosting capacity of the network is reaching its limit, the smart grid infrastructure should be ready to enable demand response (DR) programs [14,15]. The use of the smart grid to send locational marginal price (LMP) signals can improve economic efficiency in the short and long term [16]. This assumption is coherent with the regulatory provisions in the U.S, Europe, and also in Colombia, where the implantation of smart grids, distributed generation, and demand response programs are considered as key milestones in the evolution of next-generation power systems [17–19]. It is not practical for EV owners to directly bid in the

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electricity market or have transactions with electrical utilities, even they are provided with smart metering and network facilities.

A regulatory solution for the coordination of the EV connection would be the constitution of a new market actor that serves as a middleman (an aggregator) between the EV owners and the electricity market [20–22]. The EV aggregators can manage energy stored of the EV batteries to provide active power balancing as well as other ancillary services for electricity markets [23]. The debate has been focused on the allowed number of aggregators, how must be the remote control, what should be the economic signal sent to each EV connected to the grid, and the algorithms required for managing and coordinate the connection of large fleets [24–26]. A comprehensive review of above-mentioned economic and technical impacts of EV aggregators in the power system is given in [27].

Recently, Junjie et al. [28] introduced the concept of transactive control as a mean to solve the coordination problem from an economic viewpoint. In this case, the EV aggregators are exposed to uniform (uninodal) marginal prices. In general, to assure a competitive market at retail sector, it is necessary the existence of several aggregators, each one exposed to locational marginal prices (LMPs) where congestion and losses charges are distributed according to connection bus location.

However, it is important to note that ensuring a competitive market for all aggregators is mainly a regulatory issue. In this context, it is worth to evaluate what's happen when EV aggregators in a region or a country belong to only one firm. In this case, the analysis of the oligopolistic solution between a minimal cost dispatch performed by a benevolent planner and a unique EV aggregator takes relevance. The benevolent planner concerns a system operator that seeks the optimal dispatch for maximum social welfare or minimum production cost. When a company can manage a large battery capacity, its market power allows it to modify the clearing marginal prices. For this analysis, a Stackelberg solution is sought by jointly solving two optimization problems: the maximization of the profit of the EV aggregator and the optimal power flow problem. After careful review of recent contributions in this topic [20–28], we observe a lack of research on models that explicitly analyze the risks of concentrated market power on an EV aggregator. Some authors have argued that regulatory changes are needed, even when they cause undesirable effects for aggregators, they will increase market efficiency [29].

To fulfill the research gap, this paper discusses a new model that aims to evaluate -from the regulatory perspective- the economic behavior of an electric vehicle aggregator capable of exploiting business opportunities in the context of a wholesale electricity market. In particular, we investigate how the locational marginal prices (LMPs) and the 24-h power dispatch patterns are affected when only one firm acts as an EV aggregator taking advantage of the management of the storage assets. The mathematical problem is formulated as a bi-level optimization problem where the EV aggregator maximizes its profit (leader optimization problem) and the benevolent system operator performs the 24-h economic dispatch (follower optimization problem). The leader problem is solved using an evolutionary ant colony system algorithm [30], and the follower problems are solved with an optimal power flow procedure, which aims to replicate a competitive electricity market [13].

The contribution of this paper can be summarized as follows: This new approach allows us to capture the economic and technical effects of a EV aggregator regarded as a single firm in the competitive electricity market.

The paper is structured as follows: in Section 2 describes the Stackelberg model. Section 3 presents the results for two case studies: the IEEE RTS system and the Colombian power system. Then, in Section 4 it is outlined the remarks and implications of the results. Finally, Section 5 draws the conclusions of this work.

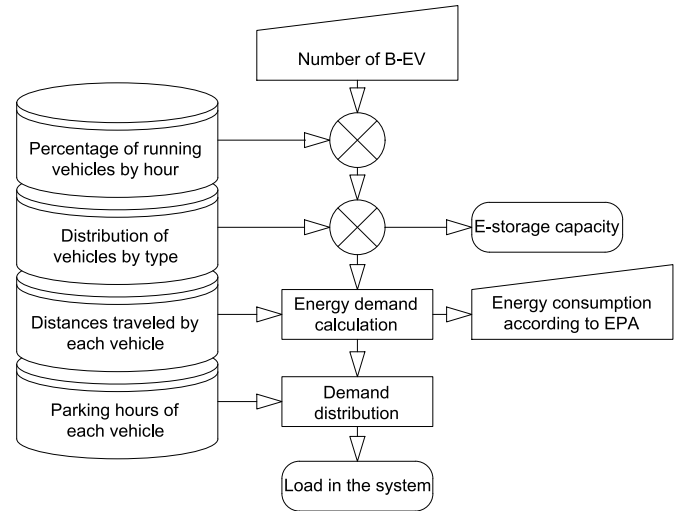


Fig. 1. Estimation flowchart for the energy demand of the EVs [33].

2. Methodology

The proposed methodology is divided into two stages: EV demand forecasting and market simulation based upon a bilevel optimization approach. The first task is devoted to forecast the daily load profile of a large fleet of EVs for a given node in the electric power system. The EV demand is added to an existing demand profile at the chosen bus. Also, in this stage, we can identify the technical and economic effects produced by the additional demand into the power system. In the second task, the EV aggregator decides -for 24-hour time window- when and where to inject/absorb EV electric power to the grid. The aggregator can manage (buy or sell) a portion of the stored energy in the EVs. The aggregator aims to maximize its benefit in an economic environment based on day-ahead marginal prices of the energy. In this part, it is possible to assess the flattening effect on the resultant demand profile.

2.1. EV Demand forecasting

In Fig. 1, a flow chart is presented for the EV demand forecasting methodology. Required input data from surveys is depicted in the left: percentage of vehicles running per hour, type and distribution of ICE vehicles, distance traveled by each ICE vehicle, and parking hours. For a specific number of EV (the size of the EV fleet), energy storage and resulting demand are determined according to the Environmental Protection Agency (EPA) of the US estimations of EV energy consumption [31,32]. The driving profiles of EVs should be similar to the same size ICE vehicles. As a result, we obtain the daily load profile and the evolution of the available e-storage capacity per hour.

In order to simplify the analysis, the fleet is characterized considering three vehicle categories: compacts, SUVs, and VANs or PickUps. So, for each category, a representative vehicle was chosen, as shown in Table 1.

The choices are justified according to the efficiency and reliability of each model. In order to determine the expected EV load profile, it is assumed that all consumed energy is refueled during the same day, and it is distributed according to the probability of the vehicle being connected to the grid. The probability depends on the time the vehicle is parked; for example, if the vehicle is parked eight hours or more, its probability of being in charging state is 1, and if the parking time is a half-hour or less, its probability is zero. Distributing the demand along the day gives a favorable scenario of uncoordinated charge with a lower impact in the daily demand profile. In contrast, the worst demand scenario will occur when all the EVs are charging at the same time with the maximum power of a DC charger 60 kW [34]. This method is appropriate to forecast the demand when the number of vehicles is high

Table 1
EVs features.

Category	Brand	Model	Year	Consumption per 100 miles [31]	E-storage capacity
Compacts	Nissan ^a	Leaf	2017	30 kWh	30 kWh
SUV	Tesla ^b	Model X AWD 90D	2017	37 kWh	90 kWh
VAN/ Pickup	Ford ^c	Transit	2012	54 kWh	28 kWh

^a <https://www.nissan-cdn.net/content/dam/Nissan/gb/brochures/>^b <https://www.tesla.com/modelx>^c <https://www.ford.com/services/assets/>

enough to apply the law of large numbers [35]. In some other approaches for small fleets, the procedure implies modeling the user's behaviors.

2.2. The Stackelberg equilibrium model

The proposed Stackelberg game is posed as a bilevel optimization problem in Eqs. (1)–(3)

$$\min_{\mathbf{x}} F(\mathbf{x}, \mathbf{L}^1, \dots, \mathbf{L}^t, \dots, \mathbf{L}^{24}) \quad (1)$$

$$\text{subject to } G_i(\mathbf{x}) \leq 0 \quad i = \{1, \dots, 3\}, \quad (2)$$

$$\begin{aligned} \mathbf{L}^t &\in \arg \min_{\mathbf{y}} \{f_i(\mathbf{y}): g_j^t(\mathbf{y}, \mathbf{x}) \leq 0 \quad j = \{1, \dots, 7\}\} \\ \forall t &= 1, \dots, 24 \end{aligned} \quad (3)$$

The function F corresponds to the objective of the leader optimization problem, defined as the maximization of the daily profit of the EV aggregator. The leader's objective depends on two sets of decision variables. The first set lists active powers exchanged during the day between the EV aggregator and the wholesale market ($\mathbf{x}$). The second set comprises daily locational marginal prices associated with each system bus ($\mathbf{L} = \mathbf{L}^1, \dots, \mathbf{L}^t, \dots, \mathbf{L}^{24}$). These prices are obtained from the solution of hourly dispatch. Profit maximization is constrained to a number energy balance functions $G_i, i = 1, \dots, 3$. At given hour t , the EV aggregator can manage a fraction of available energy stored in EV's batteries connected at its corresponding bus.

The optimal revenues of the EV aggregator not only depend on efficient energy management choices. Revenues also rely on energy prices. However, the prices depend on the market as a result of the interaction of some sellers and buyers. The market is emulated through a second optimization problem. So, for each hour t , an optimal power flow is performed to find out the economic dispatch $\mathbf{y}^t = [\mathbf{P}_G^t, \mathbf{Q}_G^t]$ that minimizes the overall social cost function f^t , subject to a number of system constraints $g_j^t, j = 1, \dots, 7$ related to global and nodal power balance as well as voltage and power flow limits. The set of locational marginal prices (LMPs) $\mathbf{L}^t \in \mathbf{L}$ at all system buses required by the leader problem are derived from Lagrange multipliers of active power balances at each bus. Details about how the LMPs are formed can be seen in [13]. It is relevant to mention that even when the aggregator has vast market power, it does not know all the information on the dispatch, only the clearing prices.

The system operator (SO) optimizes knowing the bids of the aggregator; in that sense, he becomes a follower. Decision variables for the (SO) problem at each hour t are related to generator dispatch outputs $\mathbf{y}^t$, but the set of power exchanged $\mathbf{x}^t = [\mathbf{P}_I^t, \mathbf{P}_O^t]$ is determined in the leader problem. The structure of the bilevel optimization problem is shown in Fig. 2. The solution to the OPF problem is straightforward by applying any non-linear mathematical programming procedure. The leader problem is solved by using evolutionary algorithms to simplify the search of solutions, considering that another optimization problem is being solved in the same instance. In this paper, we use the Ant Colony algorithm; details of this algorithm can be found in [30].

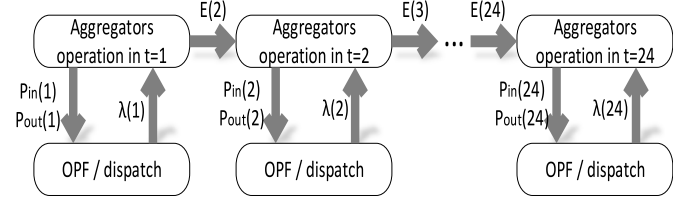
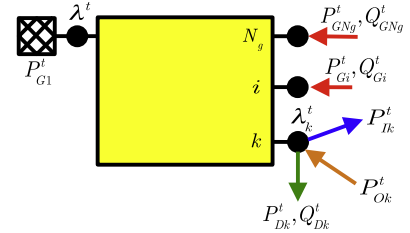


Fig. 2. Bilevel optimization approach.

Fig. 3. Representation of a EV power injection k .

2.2.1. The leader problem: maximization of the EV aggregator's profit

The EV aggregator is modeled as a prosumer, that is, a market agent able to inject/sell energy from EV's batteries for business purposes or to absorb/buy active power to charge EV's batteries for mobility purposes. Fig. 3 shows a system node $k = 1, \dots, N_a$ with a typical PQ load demand where active and reactive powers are specified. In each system node k , for each time step t , there is an EV power injection that can act either as a generator (injecting P_{Ok}^t) or as a load (consuming P_{Ik}^t). The shift from the generation state to the load state, and vice-versa will depend on locational prices and the amount of stored energy in all EV's batteries associated with node k .

The specific leader-follower (bilevel) optimization problem is now stated in a more detailed way in Eqs. (4)–(9). The EV aggregator aims to maximize the daily profit by buying or selling active power subject to daily energy balances and minimal social cost dispatch at each hour:

$$\max F(\mathbf{P}_I, \mathbf{P}_O) = \max \sum_{k=1}^{N_a} \sum_{t=1}^{24} \lambda_k^t (P_{Ok}^t - P_{Ik}^t) \quad (4)$$

subject to

$$G_1: E_k^{t+1} = E_k^t + P_{Ik}^t - P_{Ok}^t, \forall k = 1, \dots, N_a \quad (5)$$

$$G_2: \rho_k C_k^t \leq E_k^t \leq C_k^t, \forall k = 1, \dots, N_a \quad (6)$$

$$G_3: E_k^{24} = E_k^1, \forall k = 1, \dots, N_a \quad (7)$$

$$\begin{aligned} \mathbf{L}^t &\in \arg \min_{\mathbf{y}^t} f^t(\mathbf{y}^t) = \arg \min SC^t(\mathbf{P}_G^t, \mathbf{Q}_G^t) \\ \forall t &= 1, \dots, 24 \end{aligned} \quad (8)$$

Subject to:

$$g_1^t, \dots, g_7^t \text{ (network and capacity constraints)} \quad (9)$$

Daily revenues (Eq. (4)) are determined for all N_a existing nodes, where λ_k^t is the hourly locational marginal price at time $t=1, \dots, 24$ for the node k . Variable P_{Ok}^t is the power sold by the aggregator at node k ,

P_{lk}^t is the power bought by the aggregator at node k , E_k^t is the energy stored at node k and C_k^t is the total capacity of EV's batteries associated to the node k .

Three constraints are identified in the leader optimization problem. First, the optimal dispatch (P_{Ok}^{t+1} , P_{lk}^{t+1}) at hour $t + 1$ depends on the optimal dispatch at prior hour t , (P_{Ok}^t , P_{lk}^t) for all $k = 1, \dots, N_a$. This is assured by constraint G_1 (Eq. (5)). Second, the capacity parameters depend on time according to the mobility requirements of the EV owners. Thus, EV's energy to be traded in the electricity market should be restricted to a given percentage of the available capacity at time t . This verification is done by constraint G_2 (Eq. (6)). This percentage is denoted as ρ in the range from 50% to 90%. If $\rho_k = 90\%$, the aggregator is constrained to use a 10% of the available battery capacity C_k^t . A final constraint G_3 (Eq. (7)) should be included in order to ensure that the energy at the end of the day to be equal to the energy at the beginning. So, the operation is repeatable for the average day, and it is possible to estimate the expected cost for longer periods, like months or years, without increasing the optimization span.

To solve the optimization problem posed in Eqs. (4)–(9), an evolutionary algorithm programmed in Python was implemented. For each hour t , the evolutionary algorithm send last available values of P_{lk}^t and P_{Ok}^t as bids to an optimal power flow program (the follower problem).

2.2.2. Follower problem: competitive electricity market emulation

A total of 24 follower problems are defined as an economic dispatch that minimizes the overall social cost for each hour of the day. The objective is to find the active and reactive power dispatch of centralized generators that minimizes the overall production cost. The basic structure of the follower problem at hour t is given by Eqs. (10)–(17):

$$\min f^t = \min SC^t(\mathbf{P}_G^t, \mathbf{Q}_G^t) = \sum_{i=1}^{N_g} C_{Gi}(P_{Gi}^t, Q_{Gi}^t) \quad (10)$$

Subject to:

$$g_1^t: \Delta P_{losses}^t = \sum_{i=1}^{N_n} P_{Gi} - \sum_{i=1}^{N_n} P_{Di} + \sum_{k=1}^{N_a} P_{Ok} - \sum_{k=1}^{N_a} P_{lk} \quad (11)$$

$$g_2^t: P_{Gi}^t - P_{Di}^t + P_{Oi}^t - P_{li}^t = V_i^t \sum_{j=1}^{N_n} V_j^t (G_{ij} \cos \theta_{ij}^t + B_{ij} \sin \theta_{ij}^t) \quad \forall i = 1, \dots, N_n \quad (12)$$

$$g_3^t: Q_{Gi}^t - Q_{Di}^t = V_i^t \sum_{j=1}^{N_n} V_j^t (G_{ij} \sin \theta_{ij}^t - B_{ij} \cos \theta_{ij}^t), \quad \forall i = 1, \dots, N_n \quad (13)$$

$$g_4^t: F_{i,l}^t - F_{i,l}^{max} \leq 0, \quad \forall i \neq l \quad (14)$$

$$g_5^t: V_i^{min} \leq V_i^t \leq V_i^{max}, \quad \forall i = 1, \dots, N_n \quad (15)$$

$$g_6^t: P_{Gi}^{min} \leq P_{Gi}^t \leq P_{Gi}^{max}, \quad \forall i = 1, \dots, N_g \quad (16)$$

$$g_7^t: Q_{Gi}^{min} \leq Q_{Gi}^t \leq Q_{Gi}^{max}, \quad \forall i = 1, \dots, N_g \quad (17)$$

Where SC^t is the cost function at hour t , C_G is the generation cost function for active and reactive power, P_G is the real power generated, Q_G is the reactive power generated, P_D is the real power demanded, Q_D is the reactive power demanded, F is the apparent power flow, V is the voltage, θ is the voltage angle. In this model, load demands are deemed as inelastic.

We recognize seven constraints in the follower model. Constraint g_1^t (Eq. (11)) is a general active power balance equation: the sum of all injected active powers yields the system losses. Constraint g_2^t (Eq. (12)) is the nodal active power balance equation. Constraint g_3^t (Eq. (13)) is the nodal reactive power balance equation. Constraint g_4^t (Eq. (14)) is a power flow capacity equation. Voltage limits are set by constraint g_5^t (Eq. (15)). Active and reactive power limits of generators are defined in constraints g_6^t (Eq. (16)) and g_7^t (Eq. (17)), respectively.

Notice that the Lagrange multiplier associated with a given node –when $i = k$ in Eq. (12)– is just the locational marginal price λ_k^t required by the leader (Eq. (4)) to calculate the optimal revenues. When there

are no congestions in the transmission grid, a uniform (uni-nodal) price signal is sent to all buses. Specifically, for that case, λ^t corresponds to the Lagrange multiplier of the general balance Eq. (11). Furthermore, It is assumed that the marginal cost of the EV aggregator as a producer is zero, so they are always dispatched and remunerated at the corresponding nodal price. Only active power is considered in the optimization process. Additional ancillary services that the aggregator can provide are not taken into account.

Several different market players can be modelled in Eq. (10). In the following, specific cost functions are defined depending on the nature of each market player. We provide cost models for the three type of generation technologies included in the model: fossil-fired plants, non-conventional renewables (solar and wind) and large-hydro power plants.

Fossil-based market agents: The cost function of each fossil-based market agents is given by the following relationship [13]:

$$C_{Gi} = C_{zf} = \alpha_i P_{Gi}^2 + \beta_i P_{Gi} + \gamma_i \quad (18)$$

where α_i , β_i and γ_i factors will depend on the fuel type used: natural gas, coal and oil products. This function reflects costs associated with operation, maintenance, fuel and environmental charges as carbon dioxide disutilities and green certificates. Small scale distributed generators can be also represented with this cost model.

Renewable-based market agents: The cost function of each non-conventional renewable agent can be defined as [13]:

$$C_{Gi} = C_{zr} = \beta_i P_{Gi}^2 + \gamma_i \quad (19)$$

where β_i and γ_i factors will depend on the technology type used (solar, wind). The cost function only reflects expenses associated with operation and maintenance since no fossil fuel is used and therefore no environmental disutility is observed. As a result, lower operational costs in non-conventional renewables producers (solar, wind) are expected with respect to the fossil-based ones.

Large-Hydro market agents: the cost functions must be related with the opportunity cost of water [13]:

$$C_{Gi} = C_{zu} = V_Z \times T \quad (20)$$

where $V_Z = \{V_1, \dots, V_u, \dots, V_{nz}\}$, $V_u \in V_Z \subset \mathbb{R}_+$ is the effective volume of the reservoir at the end of a given period, that is T_w . This curve is interpreted as follows: if the expected volume of the reservoir u is maximum ($V_u = V_u^{max}$) the opportunity cost of water is near to zero. Conversely, if at the end of period the reservoir is at its minimum level ($V_u = V_u^{min}$) the opportunity cost of water has a predefined maximum value. At each interval $t \in T$, resulting reservoir volume V_u^t depends on three factors: the volume of water flowing through the hydro turbine qf_u , the river inflow rf_u and the spill flow sf_u [13]:

$$V_u^t = V_u^{t-1} + rf_u^t - qf_u^t - sf_u^t; \quad V_Z \times T \rightarrow \mathbb{R}_+ \quad (21)$$

Therefore, the highest volume of the reservoir u achieved at the last time interval T is given by [13]:

$$V_u^{max} = V_u^{min} + \sum_{t=1}^T rf_u^t - qf_u^t - sf_u^t; \quad V_Z \times T \rightarrow \mathbb{R}_+ \quad (22)$$

The water flow equilibrium is depicted in Fig. 4. The river inflow rf allow us to model critical scenarios where the future cost of water is an

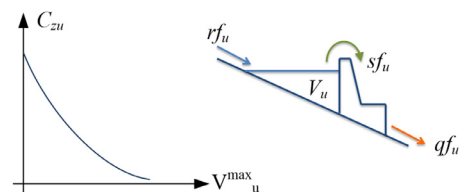


Fig. 4. Opportunity cost of water in a large reservoir dam [13].

important issue. The dispatched real power z_u of the hydroelectric plant is a function of the turbine flow q_f , that is [13]:

$$z_u = f(q_f^u) \quad \forall u \in \mathcal{U} \quad (23)$$

Each follower optimization problem is solved with PyPower optimal power flow tool [36]. The hourly economic dispatch is performed by organizing the available generators according to their costs to find the locational marginal prices of the system. The convergence criteria of the evolutionary algorithm is 0.0002 on the state variables. The convergence criteria of each OPF problem is 0.001 on the state variables.

Once the global solution of the bilevel optimization problem is reached, the results can be compared with the uncoordinated charging case estimated in Section 2.1. To do so, the valley-to-peak ratio (VPR) index is used to measure the flattening effect produced by the EV aggregator actions upon the 24-h injected power curve [19].

$$VPR_k = \frac{P_k^{t=\text{valley}}}{P_k^{t=\text{peak}}} \quad (24)$$

where the injected power is $P_k^t = P_{Gk}^t - P_{Dk}^t + P_{Ok}^t - P_{lk}^t$. The difference between the VPR ratios (with and without the effect of the EV aggregator) at given node k is expressed as a percentage in the flattening factor (FF) as follows:

$$FF_k = \left| \frac{VPR_k^{\text{with EV aggregator}} - VPR_k^{\text{without EV aggregator}}}{VPR_k^{\text{without EV aggregator}}} \right| \times 100\% \quad (25)$$

At the Stackelberg equilibrium point, the FF index can be used to characterize the overall effect of the EV aggregator on the 24-h active power curve. Note that a VPR equal to 1 will correspond to a flat 24-h power profile. A low flattening factor implies that the aggregator has a small effect upon peak reduction. A high positive, flattening factor means that the aggregator is contributing to peak reduction.

3. Case studies

The proposed methodology was applied to two case studies. The first one corresponds to the IEEE 24-bus Reliability Test System, and the second one to the Colombian power system.

3.1. IEEE 24-Bus reliability test system

The well-known IEEE 24-bus Reliability Test System is adequate to perform optimal power flow studies since it provides complete information on demand and producers cost functions [37]. All generators of the IEEE 24-bus Reliability Test System are fossil-fired based and specific cost curves (Eq. (18)) by fuel type can be found in [37]. In this paper, we consider an average weekday demand. The demand curve applied to the load buses has two peaks. The first one occurs at 11:00 a.m. (2037 MW), and the second one happens at 7:00 p.m. (2020 MW).

3.1.1. EV Demand curve estimation

In this case, the energy demand of 410,000 EVs was added. The vehicles follow the mobility patterns provided by the US NHTS survey using the methodology described in 2 [38]. On the data, vehicle distribution is the following: Compacts 52.4%, SUV 21.9%, and VAN/Pickup: 25.0%. As noted, compact cars are the most common vehicles, and they duplicate the quantity of the other categories. To incorporate the EV fleet on the IEEE RTS test system, it was divided proportionally to the original demand in the system. Therefore, there are 17 PQ load buses. The EV fleet assigned to each EV aggregator is shown in Table 2.

Fig. 5 depicts estimated 24-hour load curves for the IEEE RTS test system with and without 410,000 EVs (uncoordinated solution).

As expected, the uncoordinated inclusion of a large fleet of EVs yields to sharpen load curves. Note that a peak load with EVs is about 2500 MW, while peak load without EVs is about 2000 MW. The valley load with EVs is about 1490 MW, while peak load without EVs is about

Table 2

IEEE RTS - Vehicle distribution per bus.

Bus	Vehicles	Bus	Vehicles	Bus	Vehicles
1	15,580	7	18,040	15	45510
2	13,940	8	24,600	16	14350
3	25,830	9	25,010	18	47970
4	10,660	10	27,880	19	26240
5	10,250	13	38,130	20	18450
6	19,680	14	27,880		

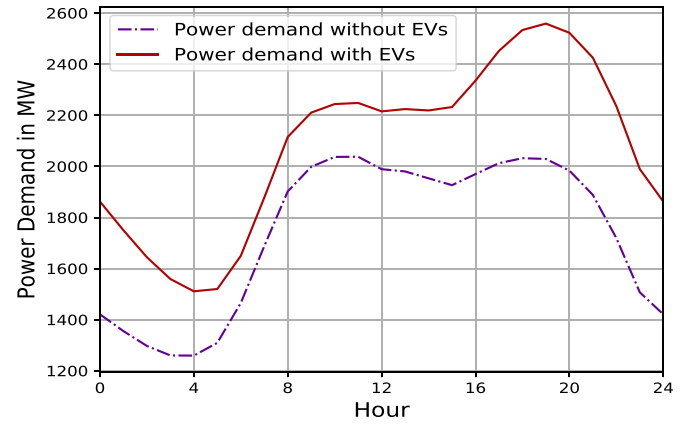


Fig. 5. IEEE RTS - uncoordinated EV demand.

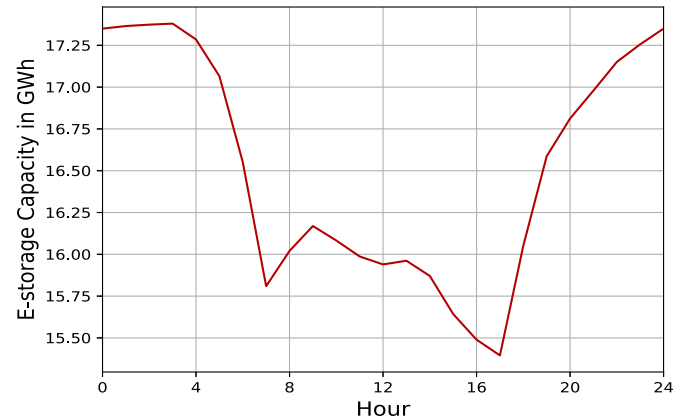


Fig. 6. IEEE RTS - available EV storage capacity.

1250 MW. Thus, the uncoordinated VPR is $1490/2500 = 0.59$. Another interesting result from the survey analysis is that there will be a huge battery capacity in the system, as shown in Fig. 6.

When roads are full (at 5:00 p.m.), there are still 88.5% of the vehicles parked. At night hours, between 12:00 p.m. and 3:00 a.m., the connection of 410,000 EVs to the grid represents a distributed battery of more than 17 GWh.

The use of probabilistic distribution functions can improve the e-storage model. In this case, the curve depicted in Fig. 6 turns from deterministic to stochastic. The proposed model is deterministic and extensions to include uncertainty are required in future works.

3.1.2. Bilevel optimization

The first case study (IEEE RTS) accounts three market players: 1) Fossil-based producers, 2) Electricity demands, and, 3) An EV aggregator.

The number of nodes with EVs is $N_a = 17$, one per PQ bus of the IEEE RTS system [39]. The EV aggregator manages an EV fleet whose size is listed in Table 2. The bilevel optimization model stated in Section 2.2 has solved assuming a battery use of 10% ($p_k = 0.9$) in all load buses.

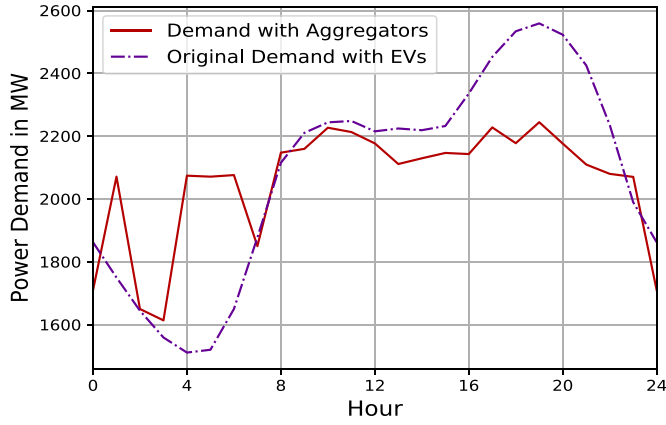


Fig. 7. IEEE RTS - 24h power profile for all system buses.

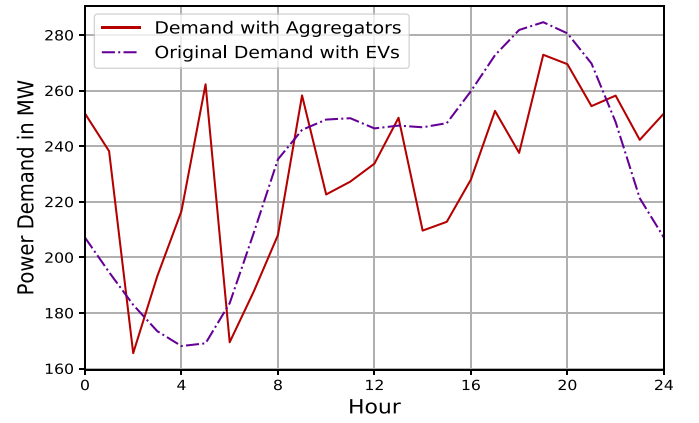


Fig. 8. IEEE RTS - 24 h power profile EV aggregator/Bus 15.

Two cases are considered. In the first one, we consider that the OPF problem is solved neglecting losses and congestion. Thus, only one uniform (uni-nodal) marginal price is determined. Later, in a second case, locational marginal prices are determined considering losses and congestion.

Case A: Uninodal marginal pricing If a uniform marginal price is applied (economic dispatch disregarding network losses and congestion), the resultant total 24-hour load curve is depicted in Fig. 7.

In this case, demand profiles at all nodes coincide with the total curve shown in Fig. 7. The VPR for the uncoordinated EV integration in all buses is known (0.59, Fig. 5). In contrast, the new VPR when the EV aggregator takes action is 0.75, which implies a flattening factor of 26.7%. As a result, the overall effect of the EV aggregator is positive. In general, the hourly marginal prices obtained are higher than those observed when the EV power injections are not included. Even if the flattening effect allows avoiding additional expansion costs on the distribution grid, the market solution is not competitive since the loss of efficiency is observed. Regarding the total daily EV aggregator income due to energy trading, it is given by:

$$\sum_{t=1}^{24} \lambda^t \sum_{k=1}^{N_a} P_{Ok}^t = \text{US\$ } 891754 \text{ per day} \quad (26)$$

The total daily EV aggregator payments for energy used to charge batteries is given by:

$$-\sum_{t=1}^{24} \lambda^t \sum_{k=1}^{N_a} P_{Ik}^t = -\text{US\$ } 130962 \text{ per day} \quad (27)$$

Then, the total profit for the aggregator is about US\$ 760,792 per day.

Case B: Locational marginal pricing approach

In this case, resultant flattening indexes are not homogeneous. Some nodes exhibit high positive revenues and resultant flat demand profiles. Other buses do not provide significant improvements, as shown in Fig. 8. The resulting curve for bus 15 has a VPR of 0.65, and its pattern is similar to the observed in the uncoordinated curve for the same bus. Therefore, the flattening factor is quite low, about 2.7%. Consequently, actions performed by the EV aggregator at node 15 do not contribute to peak reduction when locational marginal prices are applied.

Figs. 7 and 8 show steep ramps since the model allows fast charging/discharging in order to accomplish the leader economic objective (the aggregator profit). This occurs due to the existence of a single aggregator. If several actors are trading their energy in the market, the equilibrium 24-hour power curve would be smoother and more compatible with a perfect market solution as shown in [13]. A steep power ramp can affect voltage profiles and it must not be allowed in real world. Then this situation poses a regulatory issue. Future research must include ramp limits in the model.

Table 3

IEEE RTS - flattening factors per bus.

Bus	FF	Bus	FF	Bus	FF
1	12.7%	7	33.2%	15	2.7%
2	31.9%	8	12.4%	16	46.5%
3	1.1%	9	12.8%	18	5.9%
4	17.5%	10	13%	19	4.4%
5	64.3%	13	2.7%	20	48.4%
6	13.7%	14	7.4%		

Table 3 summarizes the results. Some nodes achieve an almost flat power demand. That is the case of nodes 2, 5, 7, 16, and 20, which final curves are virtually flat. Conversely, the buses with the biggest loads such as 3, 6, 13, 14, 15, 18, and 19 reached very low flattening factors since the EV aggregator does not have sufficient storage capacity to contribute to peak reduction. All nodes are contributing to reducing peak demands to some extent.

Note that despite there is a unique aggregator, each node of the system has a different price signal (LMP) and therefore a different flattening factor is achieved. This is a fundamental contribution of the paper. The aggregator must decide (at leader optimization level) according to all different LMP where, when and how much power inject from available EV storage. This is another contribution of the paper, since existing contributions are based on uniform prices and the FF are expected to be the same in all buses.

The total daily EV aggregator income due to energy trading is given by:

$$\sum_{k=1}^{N_a} \sum_{t=1}^{24} \lambda_k^t P_{Ok}^t = \text{US\$ } 629519 \text{ per day} \quad (28)$$

The total daily payments for energy used to charge batteries is given by:

$$-\sum_{k=1}^{N_a} \sum_{t=1}^{24} \lambda_k^t P_{Ik}^t = -\text{US\$ } 102.406 \text{ per day} \quad (29)$$

The EV aggregator total profit is about US\$ 527,113 per day.

Table 4 presents a summary of results. UMP is the uni-nodal marginal pricing scheme, and LMP is the locational marginal pricing

Table 4

IEEE RTS - Summary of results.

Case	Pricing Method	Profit US\$/day	Profit US\$/month per vehicle	FF
A	UMP	760,792	55.85	26.7%
B	LMP	527,113	38.52	33.7%

scheme. Results show that an EV aggregator exposed to locational marginal pricing yield lower revenues but better overall technical performance.

3.2. Colombian power system

According to [40], the Colombia power sector is structured with a number of market agents: electricity demands, large-hydro, fossil-based and wind producers. This is a hydro-dominated electricity market. Almost 64% of total power capacity and 70% of produced energy come from hydro resources, with the remainder supplied by thermal generation plants. A small share of 1% corresponds to wind producers located at La Guajira peninsula. The thermal generation is mostly gas and coal-based generation. By 2017 a total of 53 power plants were working on the system. Total electricity consumption is 48.8 TWh, which corresponds to an average energy consumption per capita of 828 kWh per year. Due lack of space cost data associated with thermal plants (Eq. (18)), non conventional renewable (solar and wind) producers (Eq. (19)) and large-hydro producers (Eq. (20)) are not included in the paper but it can be requested to the authors. Consumption per sector is divided as follows: Residential 42.2%, Industrial: 31.8%, Commercial 18%, Official 3.8%, Other uses: 4.3%. Demand is growing by approximately 4% annually. In 2008, the peak load was 9580 MW.

The proposed methodology was applied to the Colombian power system assuming an integration of 400 thousand EVs. The model is applied using day-ahead uniform (uni-nodal) marginal prices of the electricity without network losses and congestion.

3.2.1. EV Demand estimation

The expected daily load profile of Bogota including the effect of 400 and 13000 thousand EVs is shown in Fig. 9 [33]. Local regulatory agency forecasts 400 thousand EVs in Bogota by 2030 [40]. Intending to compare, we also present the energy demand of 13 million vehicles, which corresponds to the total number of particular vehicles in Colombia. It is unrealistic to assume that all of the vehicles will be electric in the short term, but the assumption that all existing vehicles would be substituted by EVs in the long term is need for comparison purposes. This example allows seeing the tendency on the load profile when the EV fleet grows. From the mobility survey of Bogota, the share of vehicle types is the following: Compacts 70.1%, SUV 16.6%, and VAN & Pickup 13.3%.

The EV storage capacity during the day for 400 thousand EVs in Colombia is shown in Fig. 10. Note that the storage capacity is averaged about 15 GW at night hours (between 9:00 p.m. and 4:00 a.m.). It is a huge amount considering that the total generation capacity of the Colombian power system is close to 16 GW. Besides, at peak hours when the streets are saturated, at 7:00 a.m. and 7:00 p.m., there are still 83% of the vehicles parked with a corresponding average capacity of 14 GW.

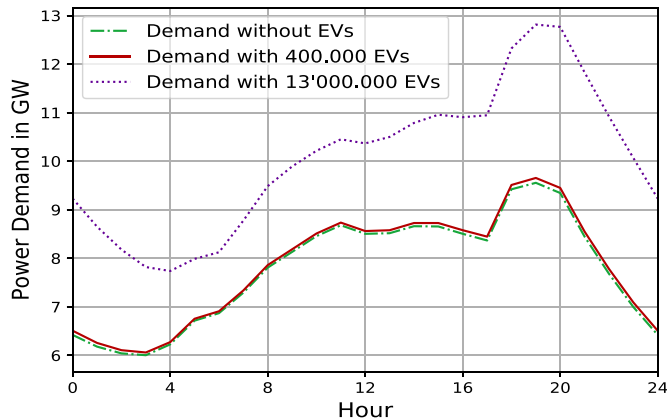


Fig. 9. Colombian electricity market - uncoordinated EV demand.

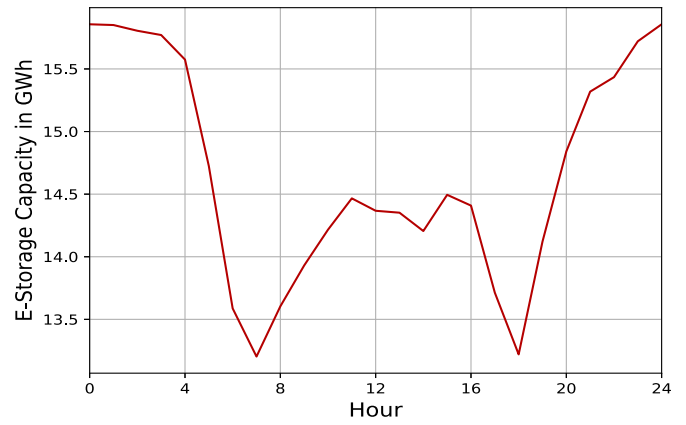


Fig. 10. Colombian electricity market - EV storage.

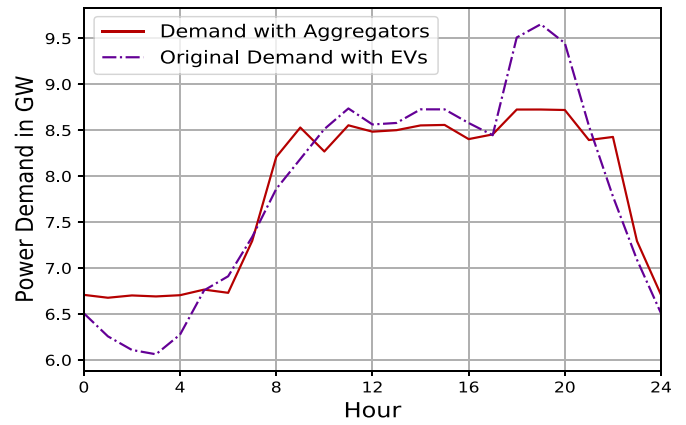


Fig. 11. Colombian electricity market - 24 h power profile with an EV aggregator.

Consequently, the EV aggregator could provide almost the same power capacity of all existing hydro plants (Fig. 11).

3.2.2. Results in the colombian power system

The second test case (the Colombian case) accounts five market players: 1) Large-hydro producers, 2) Fossil-fired producers (coal and natural gas), 3) Non conventional renewable (wind) producers, 4) Electricity demands, 5) An EV aggregator.

Colombia's electricity market is based on uniform (uni-nodal) spot prices. The cost functions of the 53 existing generators are unknown, so the dispatch order was established using the average value of the offers made by the generators during 2017 [33].

The resulting aggregate load curve with and without the EV aggregator is shown in Fig. 11. The demand with aggregator -the uncoordinated solution- has a VPR of 0.63. When the EV aggregator is included, the VPR rose to 0.76. Consequently, the flattening factor achieved is 21.8%. In this case, the EV aggregator requires to use 30% of the available capacity in batteries to achieve their maximum benefit. Turning now to the costs, with the aggregator effect, the total generation cost is 20'829,371 COP/day (1 USD = 3500 COP in 2018, Colombian currency), just 0.8% higher than the case without EVs in the system. From this total cost, the EV aggregator must pay 113'648,173 COP/day for the energy used in mobility services. Note that in the Colombian case only one flattened power curve is observed since marginal prices are the same for all buses of the system.

The total daily EV aggregator income due to energy trading is given by:

$$\sum_{t=1}^{24} \lambda^t \sum_{k=1}^{N_a} P_{Ok}^t = \text{COP\$ } 2.08 \text{ billion per day} \\ = \text{US\$ } 650920 \text{ per day} \quad (30)$$

The total daily EV aggregator payments for energy used to charge batteries is given by:

$$- \sum_{t=1}^{24} \lambda^t \sum_{k=1}^{N_a} P_{Ik}^t = - \text{COP\$ } 113 \text{ million per day} \\ = - \text{US\$ } 35515 \text{ per day} \quad (31)$$

The EV aggregator total profit is about US\$ 615,405 per day.

4. Technical and market implications of the EV aggregation

In this research, we found that the EV aggregator does not require a significant share of batteries' capacity to achieve the best profit outcome. As a key result, we show that when the EV aggregator tries to achieve their optimal profit in the wholesale market resulting power curves tend to be more flattened. This occurs because EV vehicles are injecting power into the grid at peak hours, avoiding line congestion and the dispatch of more expensive centralized plants. Thus, in the account of huge e-storage capacity, the EV aggregator can be able to balance the system by participating in the wholesale electricity market as an energy trader and flattening the load curve as a collateral effect. However, the Stackelberg solution produces higher marginal prices in the follower problems. Thus, some loss of social welfare is observed. These societal costs could be compensated to some extent with the avoided costs associated with network expansion.

5. Conclusions

This paper presents an economic model that aims to evaluate -from the regulatory perspective- the behavior of an aggregator capable of exploiting business opportunities of electric vehicles (EV) in the context of the competitive electricity market.

A Stackelberg game is formulated as a bi-level optimization problem where the maximization of the EV aggregator profit is regarded as the leader's optimization problem, and the set of hourly economic dispatches and the corresponding locational marginal prices (LMPs) are determined by a benevolent system operator (the follower's optimization problem).

At the equilibrium point, the resulting 24-h real power curves show important peak reductions with respect to the solution without the inclusion of the EV aggregator. In the long-term, this situation might be a good coordination solution for a large EV fleet allowing to avoid investments in grid expansion. However, in the short-term, the Stackelberg solution yields some loss of economic efficiency being far from the competitive solution provided by several different EV aggregators.

Results obtained from the RTS IEEE test case show different flattened power curves since resulting locational marginal prices are different at each bus. In the Colombian case only one flattened power curve is observed since marginal prices are the same for all buses of the system. In both cases, it is observed that the EV aggregator only requires a low share of existing storage capacity (10%) to achieve the a good profit outcome.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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